NGUYEN HAI NAM

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EDUCATION

Hanoi University of Science and Technology | Hanoi, Vietnam

Bachelor of Computer Science *(Expected Graduation: June 2027)* **2023 – Present**

Relevant Coursework: Data Structures & Algorithms, Database Systems, Computer Systems, Machine Learning, Data Mining, Natural Language Processing, Probability & Statistics

PROJECTS

OnTrade | Automated End-to-End Quantitative Trading System **April 2026 – June 2026**

Quant Researcher & Data Engineer | github.com/namh1012rj-sketch/OnTrade-Quant-System

- Designed & implemented an ETL data pipeline in Python using **SQLAlchemy & PostgreSQL** to clean & ingest 5+ years of market OHLCV & financial data for VN50.
- Engineered high-performance Backtesting Engine core in **C++17 via Ctypes**, achieving an **80x speedup** (reduced simulation runtime from 12s to 0.15s).
- Developed a 50+ factor Hybrid Feature Framework (Technical & Fundamental) and trained **XGBoost Classifier** with walk-forward validation (68.7% accuracy, Sharpe 1.50).
- Achieved **+38.53% Total Return** over 10-month out-of-sample testing while maintaining conservative risk control (Max Drawdown -8.12%, Profit Factor 8.02).

DMX AI Agent | Intelligent E-Commerce Shopping Assistant **July 2026 – August 2026**

AI & Data Engineer | github.com/namh1012rj-sketch/-i-n-m-y-xanh-chatbot-

- Architected & built a **ReAct Agent** and **RAG system** using LLMs (DeepSeek/Gemini) and multilingual-e5-large embeddings for automated product recommendations.
- Integrated **MongoDB Atlas Vector Search (HNSW)** with Cosine Similarity for intelligent semantic query retrieval, eliminating keyword matching limitations.
- Optimized retrieval pipeline with multi-threading (**ThreadPoolExecutor**) & thread-safe caching, accelerating vector query speed by **~6.8x**.
- Engineered streaming response pipeline supporting **Server-Sent Events (SSE)** for real-time output and sliding-window conversation memory management.
- Developed an automated prompt optimization system (**auto_optimizer.py**) with LLM-as-a-Judge evaluation and dynamic budget tolerance filtering (+15%).

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI / Machine Learning: ReAct Framework, RAG Systems, Vector Search (HNSW), Sentence-Transformers / Embeddings, LLM APIs (DeepSeek, Gemini, OpenAI), XGBoost, LightGBM, Pandas, NumPy

Databases & Tools: MongoDB Atlas, PostgreSQL, SQLAlchemy, Multi-threading / Concurrency, Git, Docker, Pytest, Ctypes, Linux/Bash, VS Code

AWARDS & CERTIFICATIONS

Certifications: IELTS 6.5

Competitions: ICPC Vietnam National Programming Contest – Ranked 82nd | Vietnam AI Hackathon 2026 – Participant